

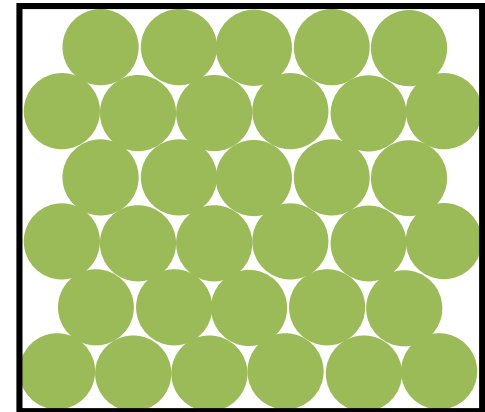
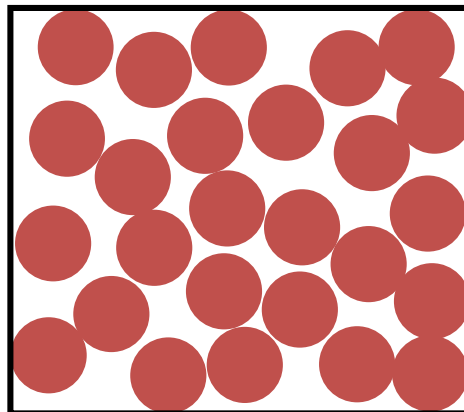
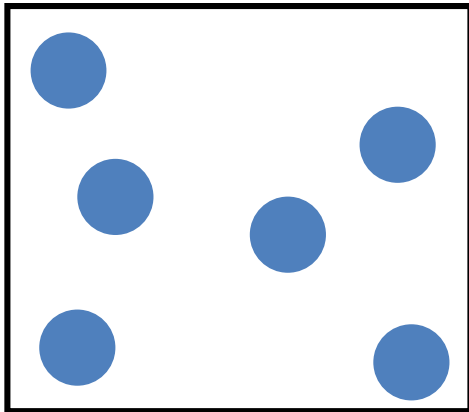
Physical Chemistry of Materials

Lecture 10

Solids

Gas, Liquid, and Solid

condensed phase



Gas phase: kinetic energy $\gg$ potential energy

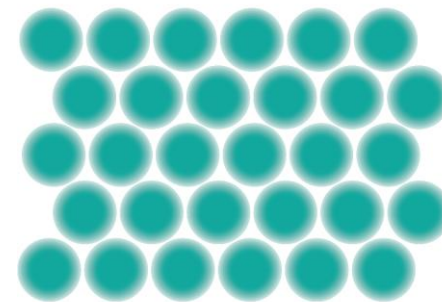
Condensed phase: kinetic energy $\approx$ potential energy

$$\frac{1}{2}mv^2$$

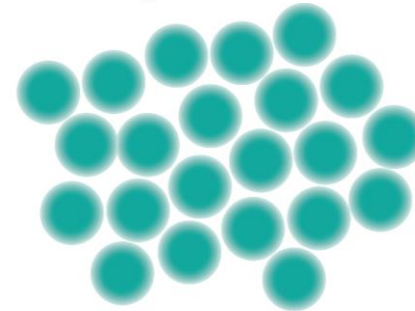
various forms

Condensed Phases

- Crystalline solids
- Amorphous solids
- Liquids (cf. glasses)
- “Soft matters”
 - Polymers
 - Colloids
 - Liquid crystals ...



Crystalline



Amorphous

소재물리화학의 구성

고전 열역학

열역학 복습

상 전이

통계역학

기초 이론

이상 기체

상호작용 계

고체

고체의 구조

고체의 진동

고체의 전기적 성질

고체의 자기적 성질

표면화학

나노화학

연성물질

액체

유리

고분자의 구조

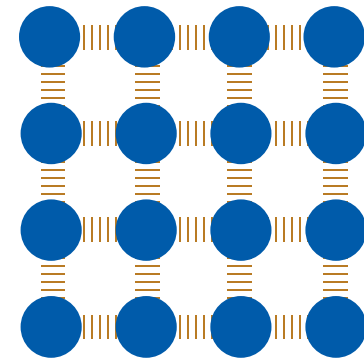
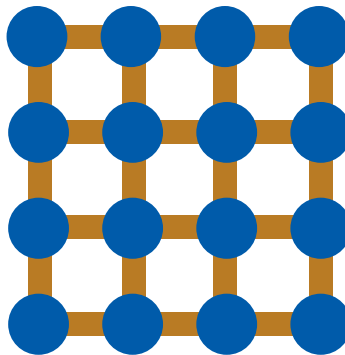
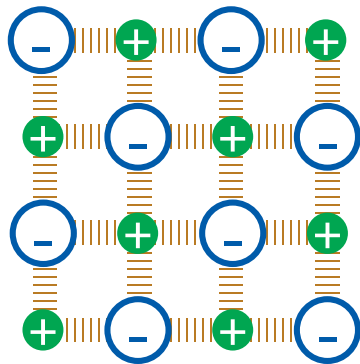
고분자의 상 거동

고분자의 젤화

액정, 콜로이드, ...

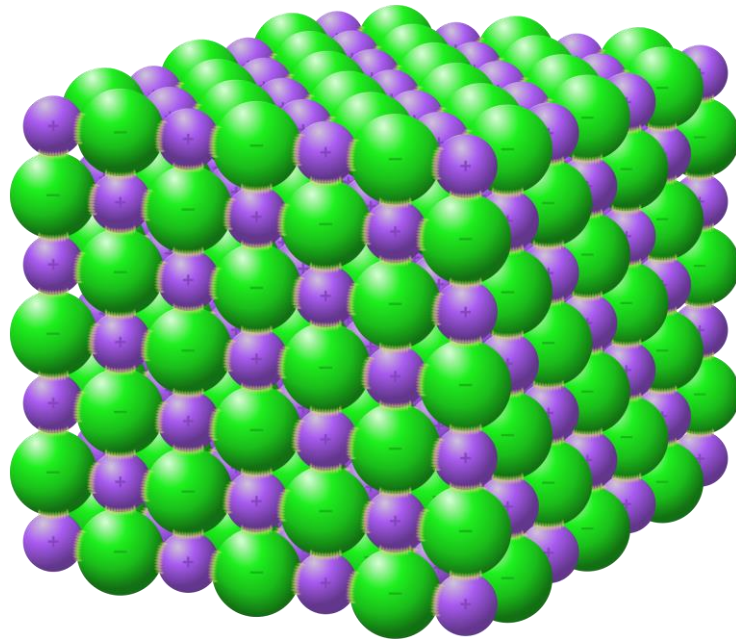
Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions



Metals need more discussion

Ionic Solids



Sodium chloride (NaCl)

(Image: https://commons.wikimedia.org/wiki/File:NaCl_bonds.svg)

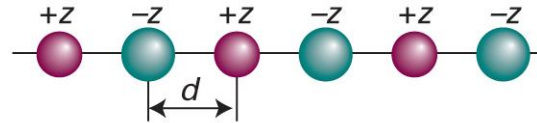
Lattice Energy of Ionic Solids

- Lattice energy:



- Two main contributions
 - Electrostatic interactions (attraction + repulsion)
 - Pauli repulsion

1-Dimensional Model



$$\begin{aligned} E_p &= \frac{1}{4\pi\epsilon_0} \left(-\frac{z^2 e^2}{d} + \frac{z^2 e^2}{2d} - \frac{z^2 e^2}{3d} + \frac{z^2 e^2}{4d} + \dots \right) \\ &= \frac{z^2 e^2}{4\pi\epsilon_0 d} \left(-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \dots \right) = -\frac{z^2 e^2}{4\pi\epsilon_0 d} \times \ln 2 \end{aligned}$$

Consider two sides for each ion:

$$E_p = -2 \ln 2 \times \frac{z^2 e^2}{4\pi\epsilon_0 d}$$

Generalization

$$E_p = -2 \ln 2 \times \frac{z^2 e^2}{4\pi\epsilon_0 d}$$

For ions with arbitrary charges z_A and z_B in 3 dimensions,

$$E_p = -A \times \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d}$$

where A is the Madelung constant (structure-dependent).

Pauli Repulsion

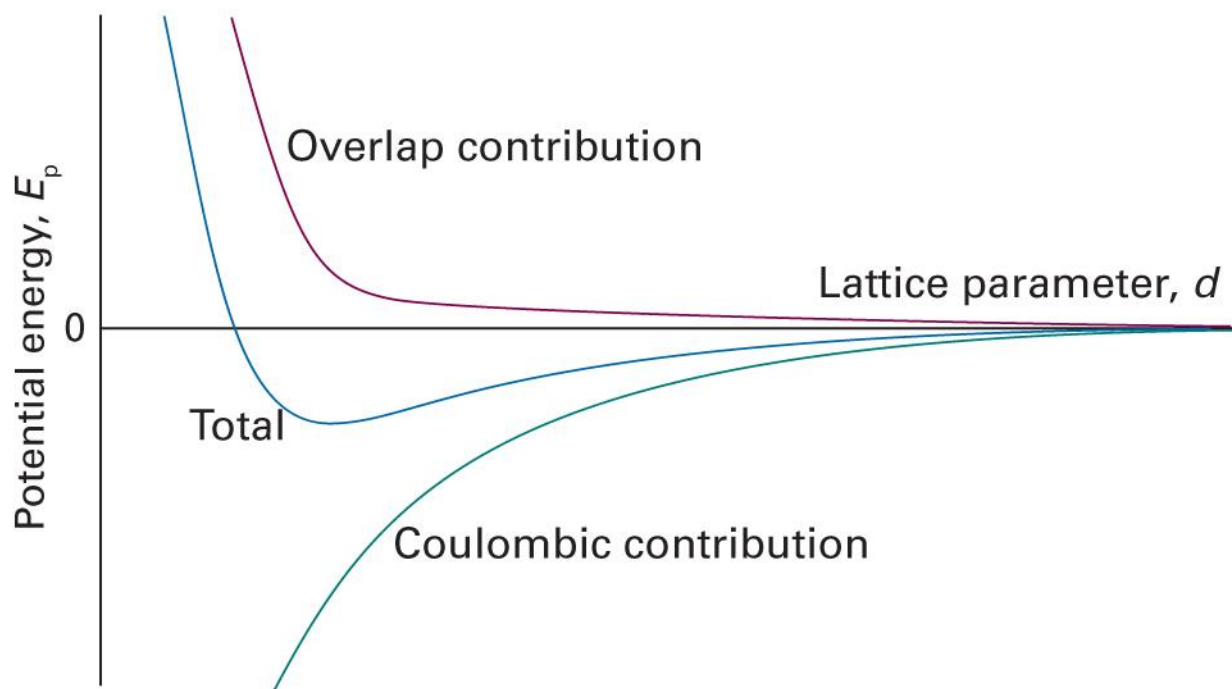
modeled as an exponential potential (lecture 07)

$$E_p^* = C e^{-d/d^*}$$

- Total potential energy $E_{p,\text{tot}} = E_p + E_p^*$

$$E_{p,\text{tot}} = -A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d} + C e^{-d/d^*}$$

Total Potential Energy



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 15C.13)

Determination of d_{eq}

$$E_{\text{p,tot}}(d) = -A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d} + C e^{-d/d^*}$$

The structure is stabilized when $E_{\text{p,tot}}$ is minimal:

$$\frac{dE_{\text{p,tot}}}{dd} = A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}^2} - \frac{C}{d^*} e^{-d_{\text{eq}}/d^*} = 0$$

$$A d^* \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}^2} = C e^{-d_{\text{eq}}/d^*}$$

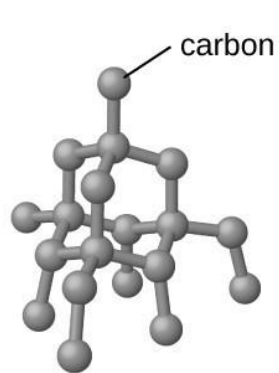
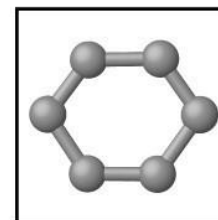
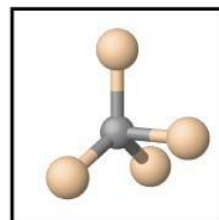
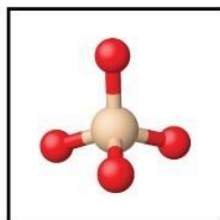
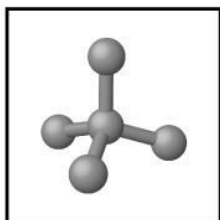
Born-Mayer Equation

$$Ad^* \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}^2} = C e^{-d_{\text{eq}}/d^*}$$

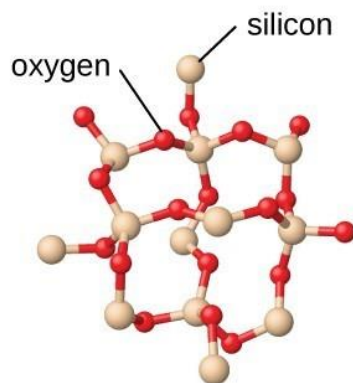
$$\begin{aligned} E_{\text{p,tot}}(d_{\text{eq}}) &= -A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}} + C e^{-d_{\text{eq}}/d^*} \\ &= -A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}} + Ad^* \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}^2} \end{aligned}$$

$$E_{\text{p,tot}}(d_{\text{eq}}) = -A \frac{|z_A z_B| e^2}{4\pi\epsilon_0 d_{\text{eq}}} \left(1 - \frac{d^*}{d_{\text{eq}}} \right)$$

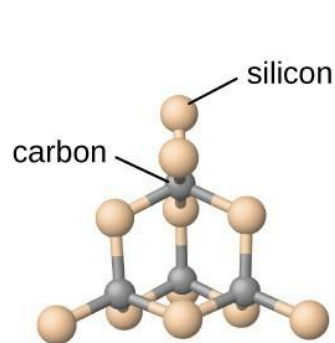
Covalent Solids



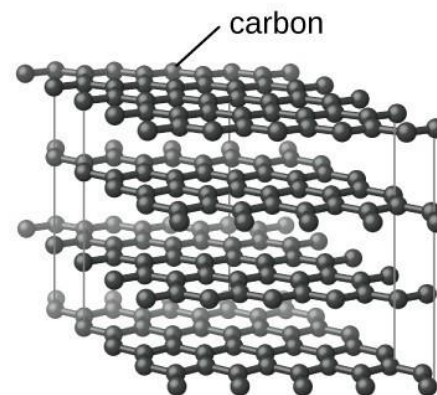
diamond



silicon dioxide

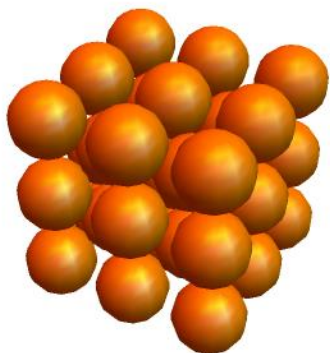


silicon carbide

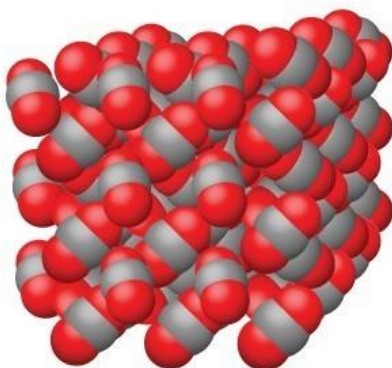


graphite

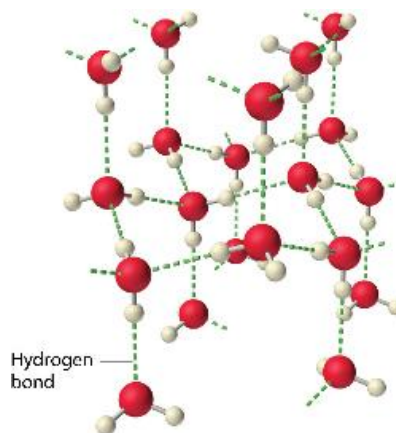
Molecular Solids



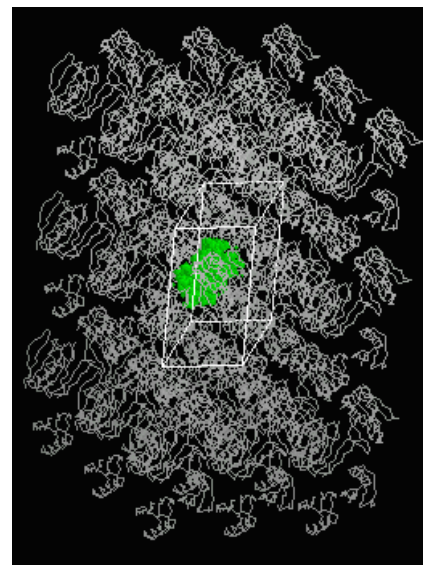
argon



carbon dioxide



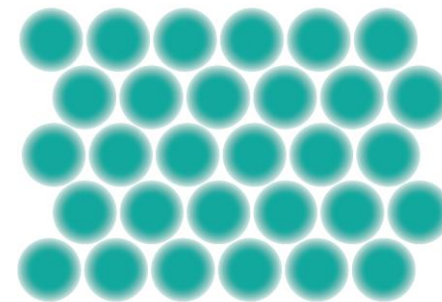
water



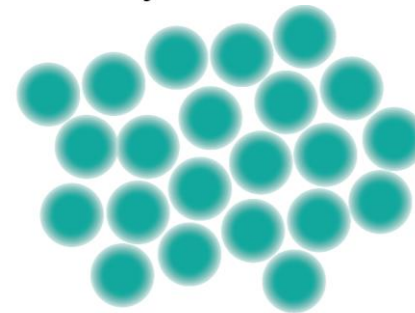
protein

Condensed Phases

- Crystalline solids ←
- Amorphous solids
- Liquids (cf. glasses)
- “Soft matters”
 - Polymers
 - Colloids
 - Liquid crystals ...



Crystalline



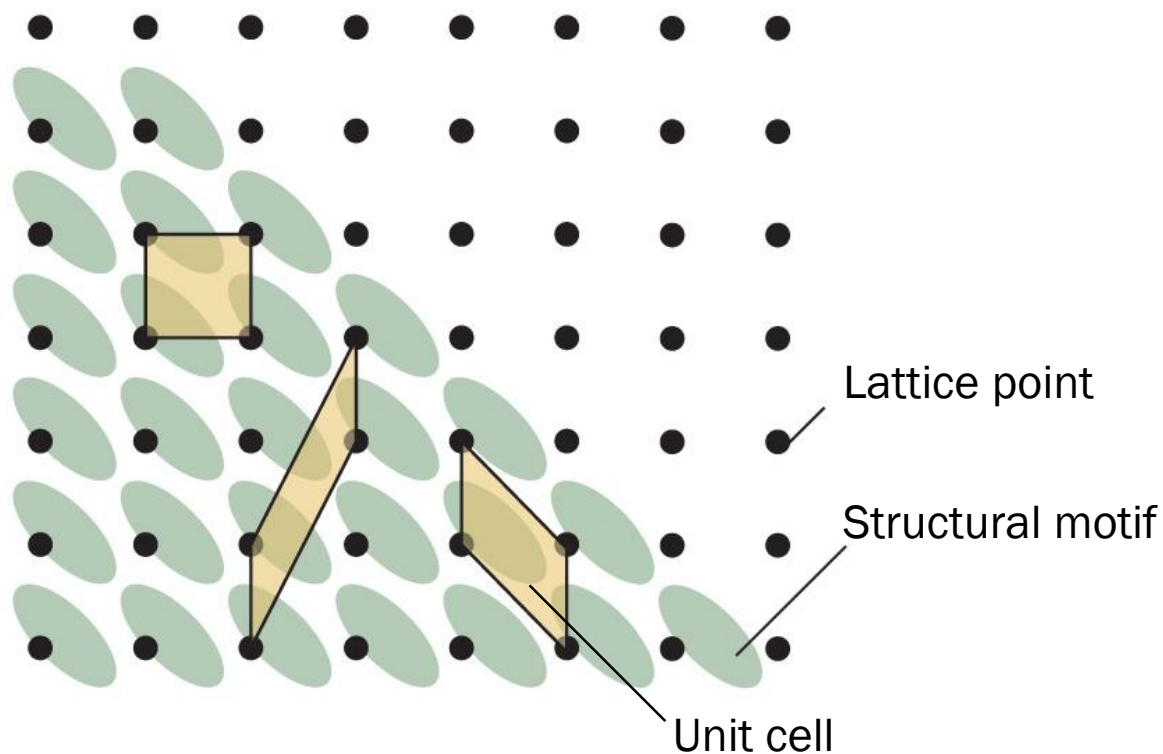
Amorphous

Crystalline Solids

- Perfect crystals: built up from regularly repeating “structural motifs”
- Real crystals can contain defects

Unit Cells

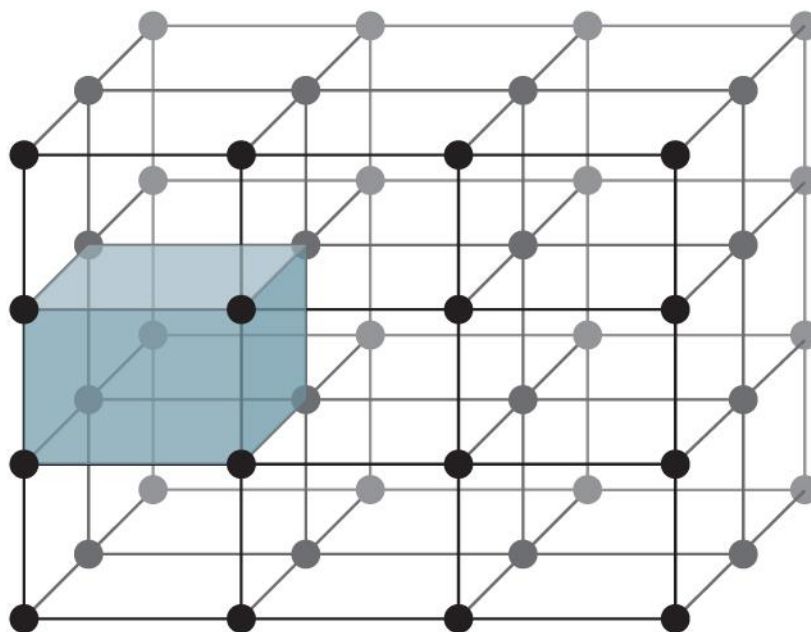
whose purely translational displacements can span the entire space



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 15A.1 and 15A.2)

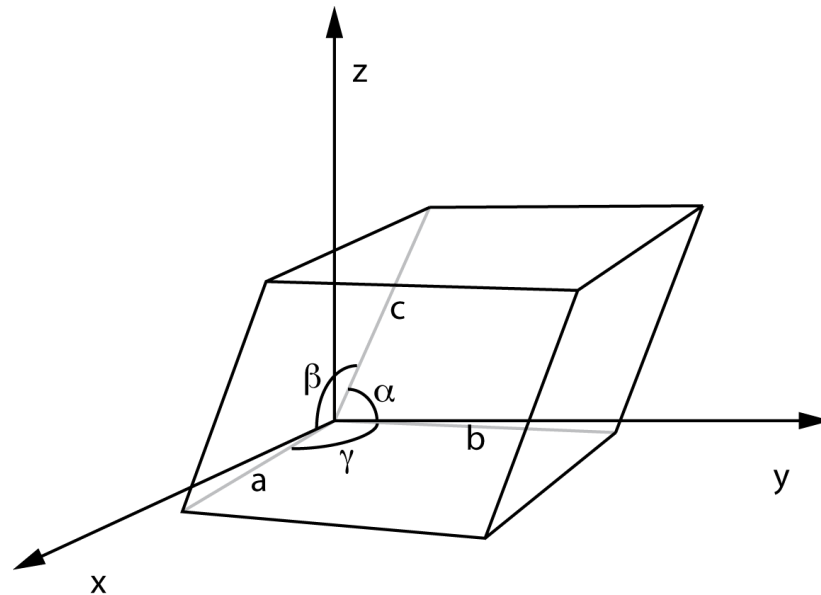
Primitive Unit Cell

contains **only one** lattice point



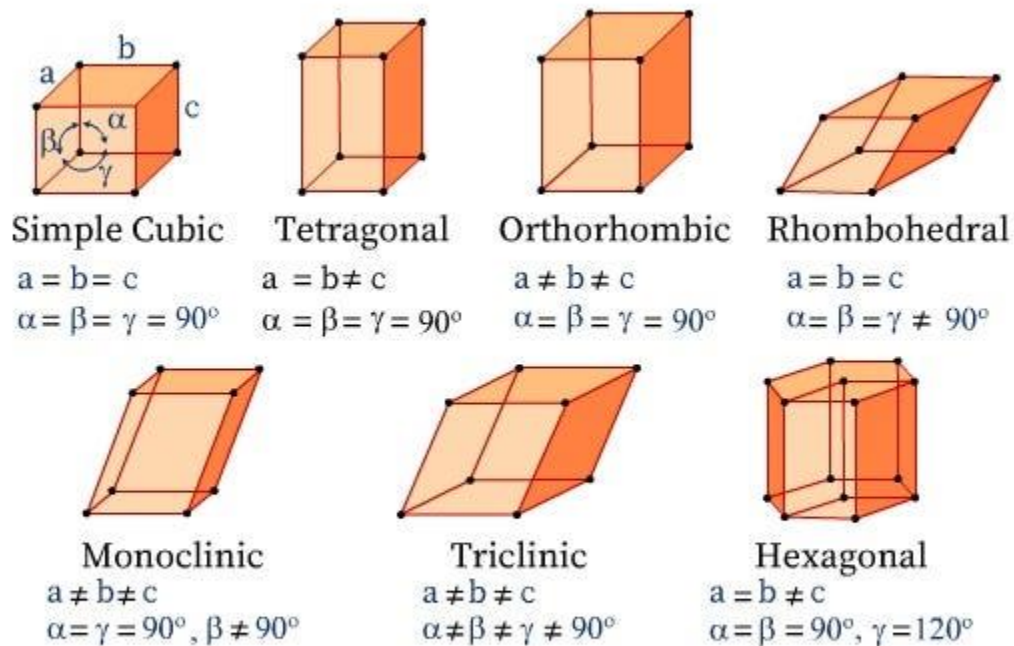
Lattice Constants

- Lengths of sides: a, b, c
- Angles between sides: α, β, γ

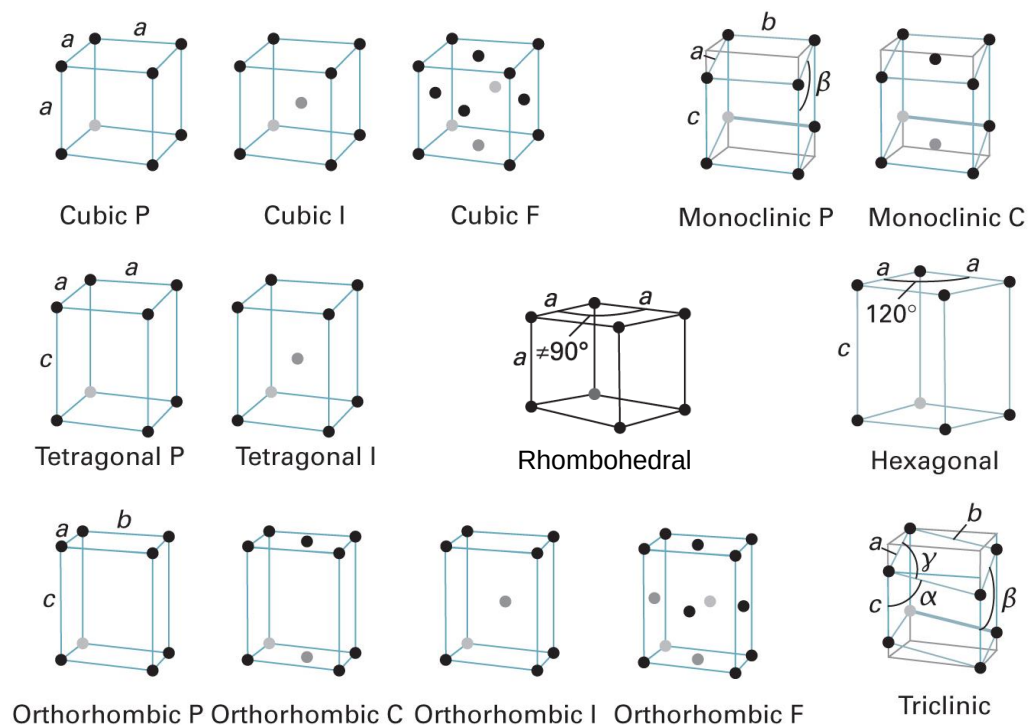


(Image: <https://commons.wikimedia.org/wiki/File:UnitCell.png>)

Lattice Systems



14 Bravais Lattices

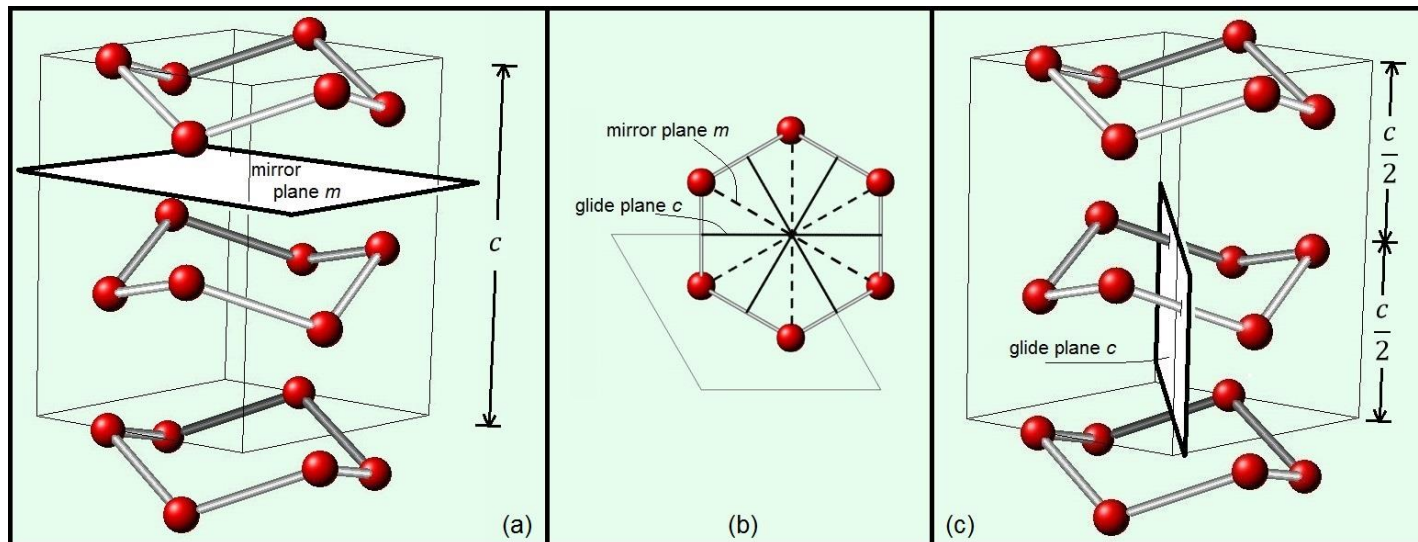


(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 15A.8)

Crystal Symmetry

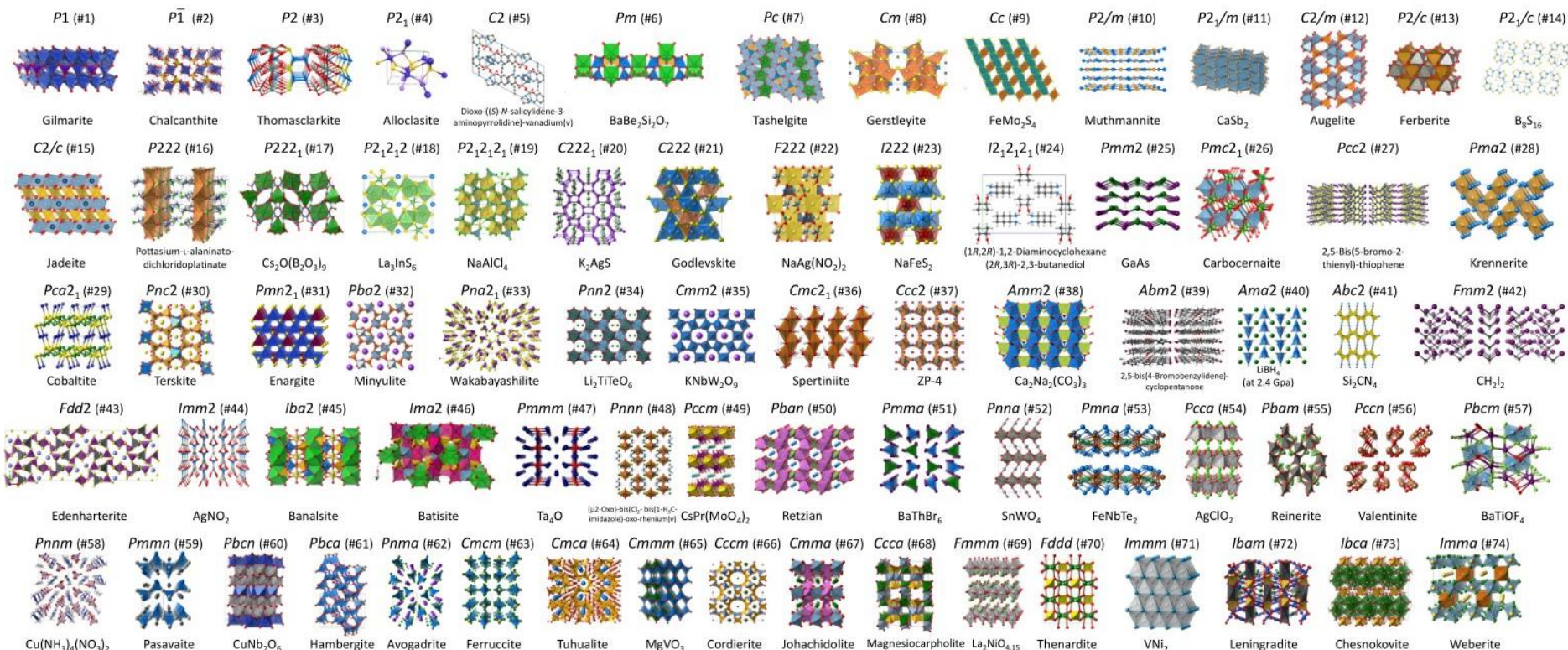
lattice: hexagonal

Ice I_h system

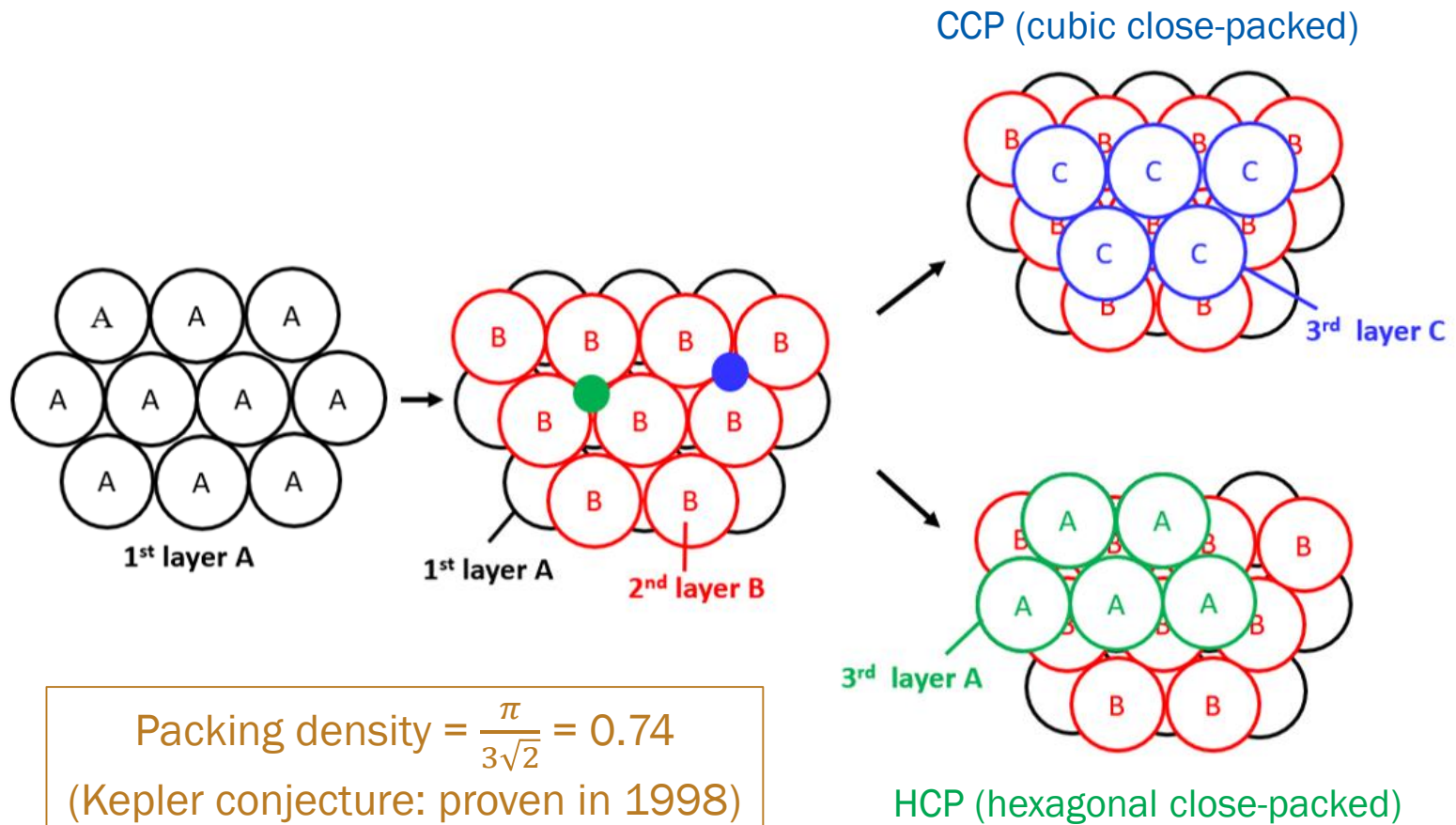


lattice symmetry $\neq$ crystal symmetry

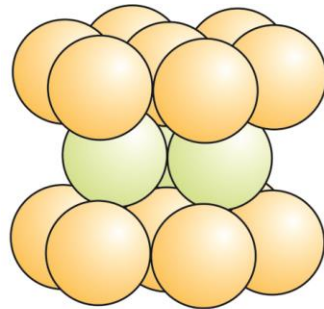
230 Space Groups



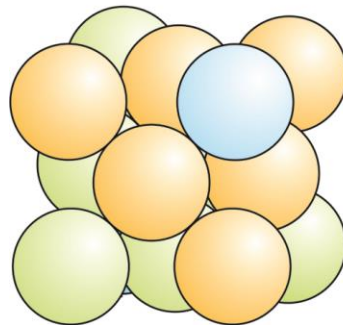
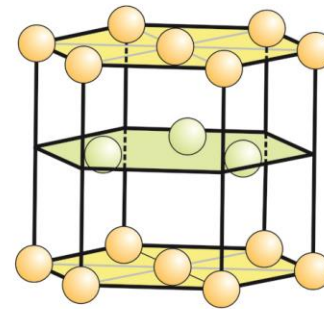
Close Packing



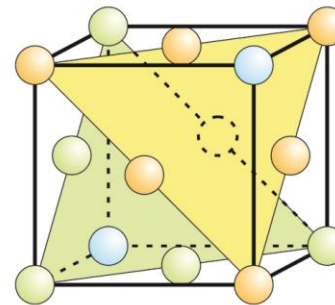
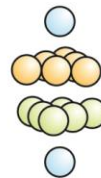
Unit Cells of CCP and HCP



a. hexagonal closest packing



b. cubic closest packing



Crystal Structures of Elements at STP

STP - Standard Temperature and Pressure

Crystal Structures of Elements at STP																	He					
STP - Standard Temperature and Pressure																	HCP					
H HEX																						
Li BCC	Be HCP	BCC - Body-centered Cubic FCC - Face-centered Cubic HEX - Simple Hexagonal HCP - Close-packed Hexagonal DHCP - Double Close-packed Hexagonal RHO - Rhombohedral										BCT - Body-centered Tetragonal ORTH - Orthorhombic DC - Diamond Cubic DT - Diamond Tetragonal SC - Simple Cubic * predicted crystal structure					B RHO	C HEX	N complex HCP	O P-cubic	F P-cubic	Ne FCC
Na BCC	Mg HCP																Al FCC	Si DC	P ORTH	S ORTH	Cl complex C-ORTH	Ar FCC
K BCC	Ca FCC	Sc HCP	Ti HCP	V BCC	Cr BCC	Mn α-Mn	Fe BCC	Co HCP	Ni FCC	Cu FCC	Zn HCP	Ga complex F-ORTH	Ge DC	As P-RHO	Se complex HEX	Br complex C-ORTH	Kr FCC					
Rb BCC	Sr FCC	Y HCP	Zr HCP	Nb BCC	Mo BCC	Tc HCP	Ru HCP	Rh FCC	Pd FCC	Ag FCC	Cd HCP	In BCT	Sn DT	Sb P-RHO	Te complex HEX	I complex C-ORTH	Xe FCC					
Cs BCC	Ba BCC	57-71	Hf HCP	Ta BCC	W BCC	Re HCP	Os HCP	Ir FCC	Pt FCC	Au FCC	Hg RHO	Tl HCP	Pb FCC	Bi RHO	Po SC	At FCC*	Rn FCC*					
Fr BCC*	Ra BCC	89-103	Rf HCP*	Db BCC*	Sg BCC*	Bh HCP*	Hs HCP*	Mt FCC*	Ds BCC*	Rg BCC*	Cn HCP*	Nh HCP*	Fl FCC*	Mc UNKNOWN	Lv UNKNOWN	Ts UNKNOWN	Og FCC*					



Solid state at STP



Liquid state at STP

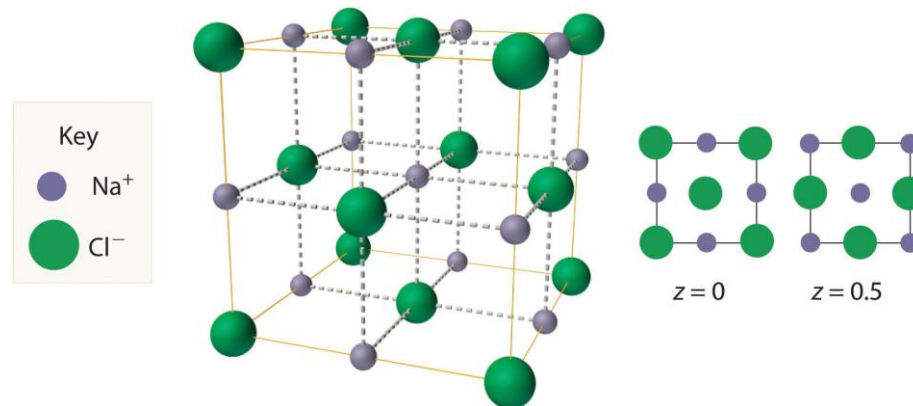


Gaseous state at STP

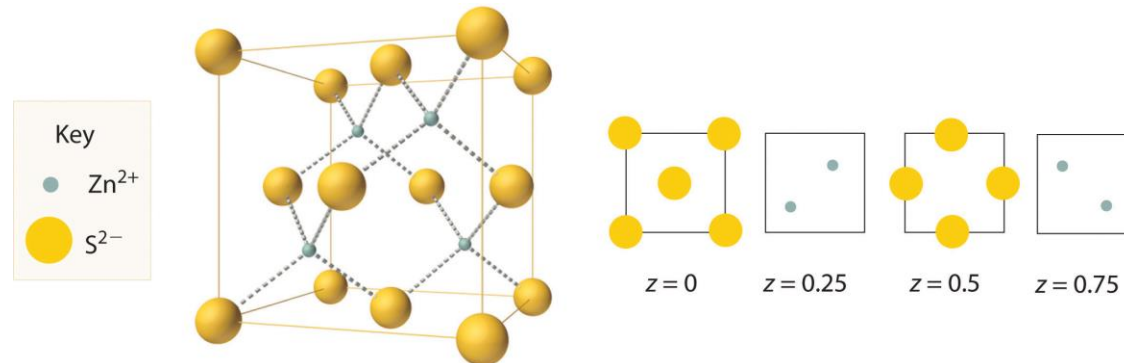
La DHCP	Ce DHCP	Pr DHCP	Nd DHCP	Pm DHCP	Sm complex RHO	Eu BCC	Gd HCP	Tb HCP	Dy HCP	Ho HCP	Er HCP	Tm HCP	Yb FCC	Lu HCP
Ac FCC	Th FCC	Pa BCT	U ORTH	Np ORTH	Pu MONO	Am DHCP	Cm DHCP	Bk DHCP	Cf DHCP	Es FCC	Fm FCC*	Md FCC*	No FCC*	Lr HCP*

Binary Compounds

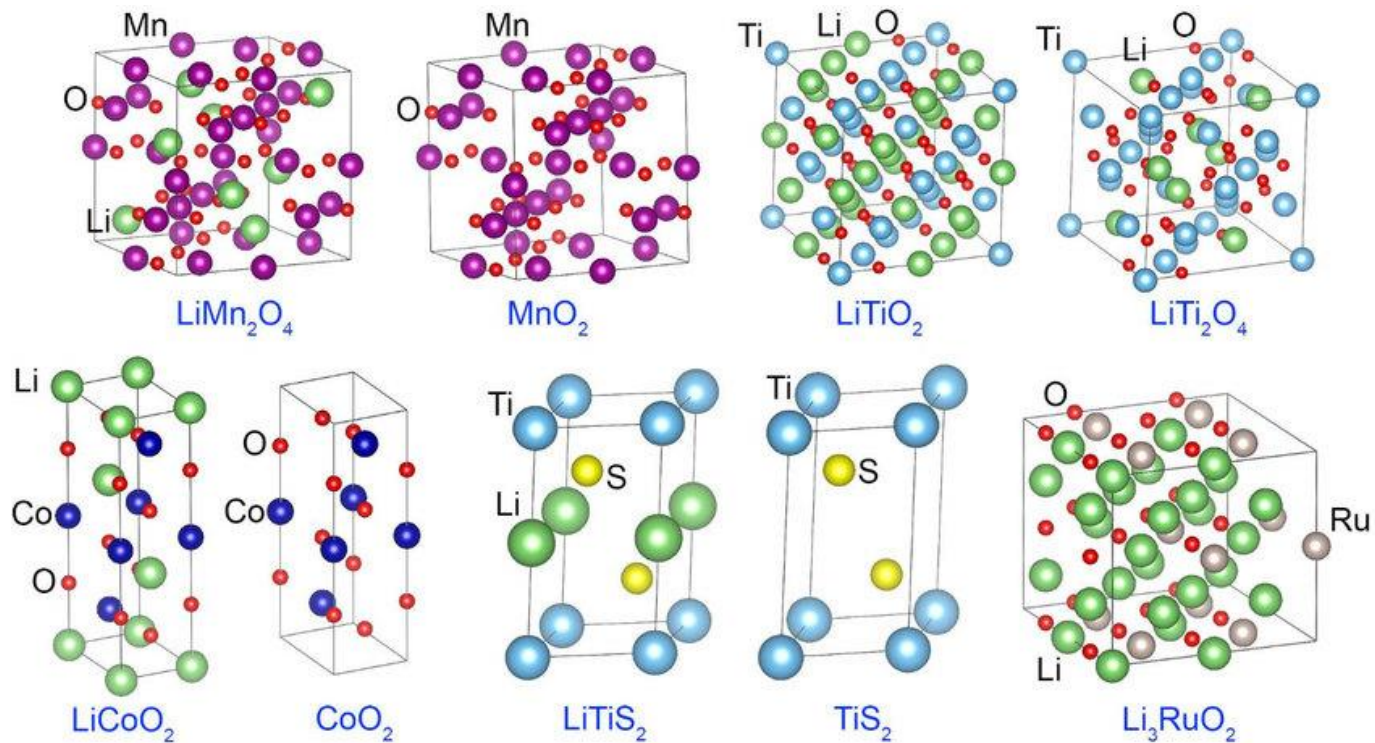
Rock-salt
structure



zinc-blend
structure

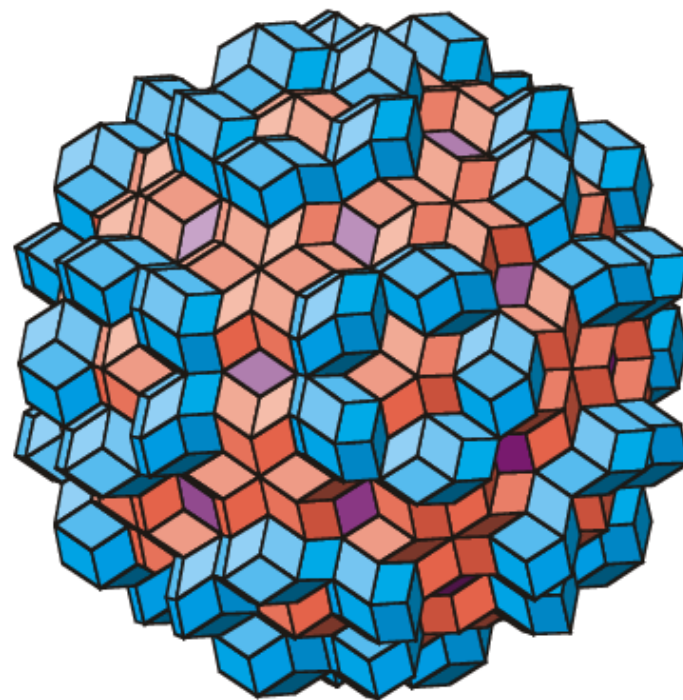
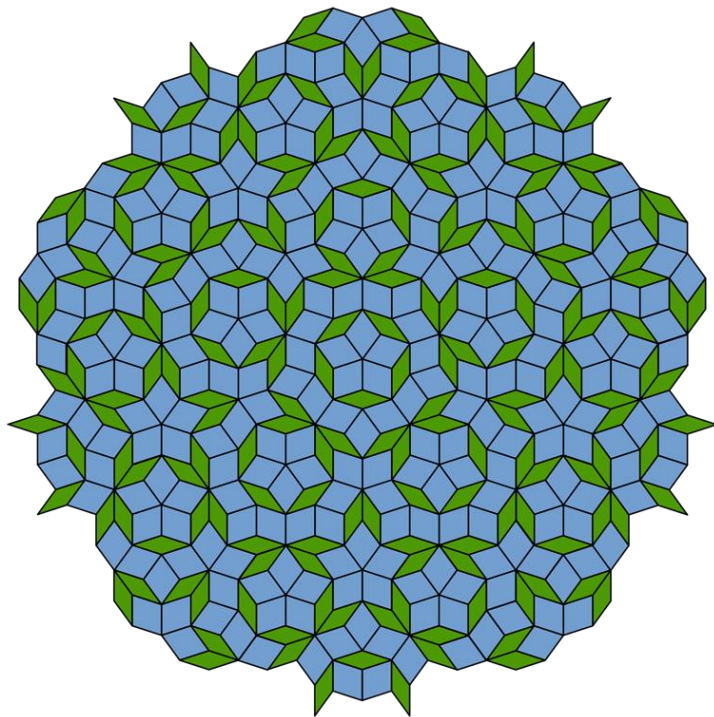


Getting Complicated ...



(Image: <https://doi.org/10.48550/arXiv.1512.08270>)

Quasicrystals

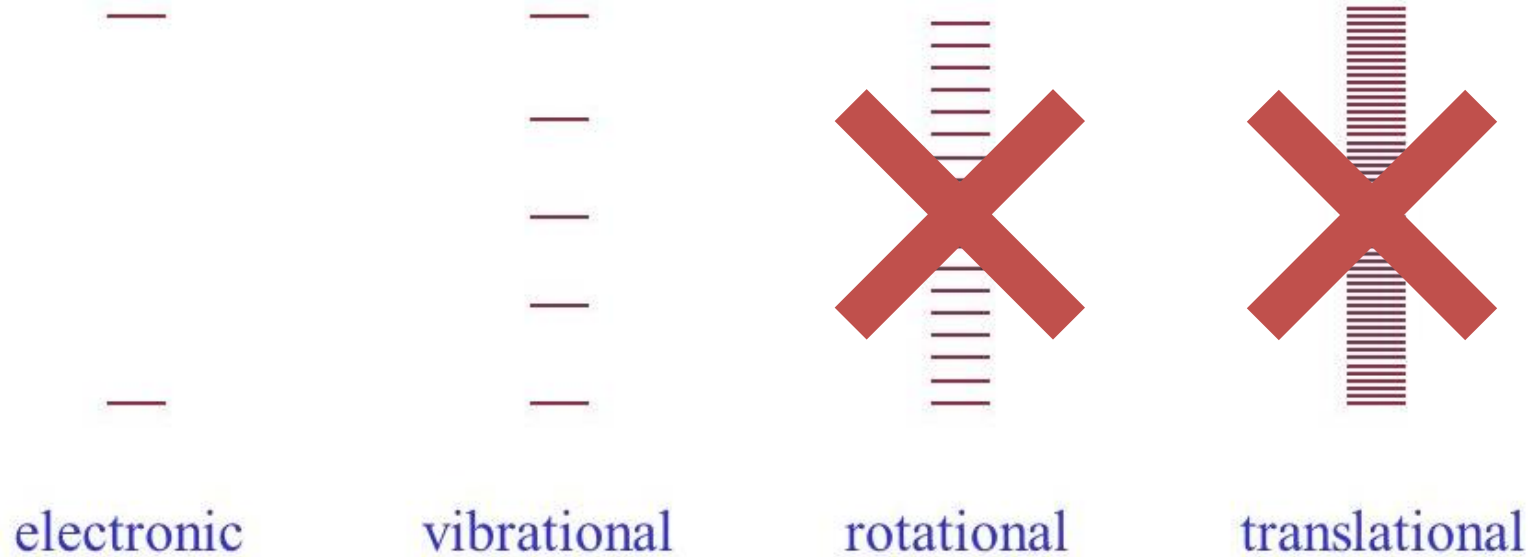


Nobel prize in Chemistry 2011

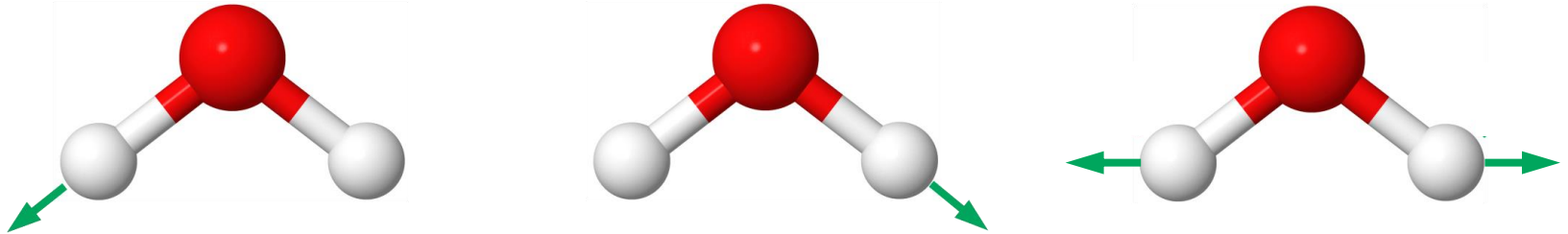
(Image: <https://physics.aps.org/story/v28/st14> & Madison, *RSC Advances* 2015)

Crystal Energies

Molecular case:



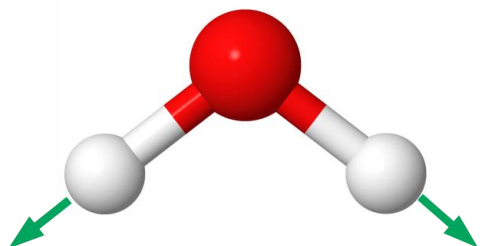
Molecular Vibration



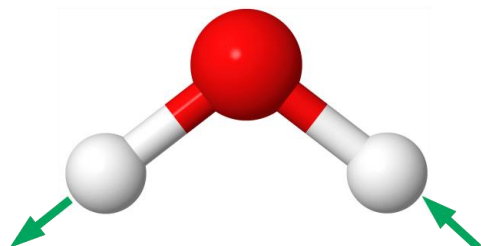
Center of mass is not fixed!
(coupled with translation)

Normal Modes

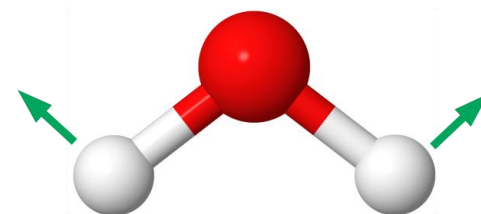
a pattern of motion in which all atoms move with the same frequency



symmetric stretching

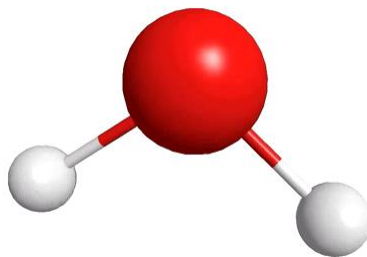


asymmetric stretching



bending

3551 cm⁻¹



Taylor Expansion

For an arbitrary potential energy

$$\begin{aligned} V(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) &= V(x_1, y_1, z_1, x_2, y_2, z_2, \dots, x_N, y_N, z_N) \\ &= V(q_1, q_2, q_3, q_4, q_5, q_6, \dots, q_{3N}), \end{aligned}$$

if the system is close to its equilibrium structure ($q_i = 0$),

$$\begin{aligned} V(\{q_i\}) &= \cancel{V_0} + \sum_{i=1}^{3N} \left(\cancel{\frac{\partial V}{\partial q_i}} \right)_0 q_i \\ &\quad + \frac{1}{2} \sum_{i=1}^{3N} \left[\left(\frac{\partial^2 V}{\partial q_i^2} \right)_0 \right]^{k_{ii}} q_i^2 + \frac{1}{2} \sum_{i=1}^{3N} \sum_{j>i}^{3N} \left[\left(\frac{\partial^2 V}{\partial q_i \partial q_j} \right)_0 \right]^{2k_{ij}} q_i q_j + \dots \end{aligned}$$

(Lots of) Harmonic Oscillators

$$V = \frac{1}{2} \sum_{i=1}^{3N} k_{ii} q_i^2 + \frac{1}{2} \sum_{i=1}^{3N} \sum_{j \neq i}^{3N} k_{ij} q_i q_j = \frac{1}{2} \sum_{i=1}^{3N} \sum_{j=1}^{3N} k_{ij} q_i q_j$$

The force along coordinate i is

$$F_i = -\frac{\partial V}{\partial q_i} = -\sum_j k_{ij} q_j = m_i a_i = m_i \frac{\partial^2 q_i}{\partial t^2}$$

$$\frac{\partial^2 q_i}{\partial t^2} + \frac{1}{m_i} \sum_j k_{ij} q_j = 0$$

Coherent Movement

$$\frac{\partial^2 q_i}{\partial t^2} + \frac{1}{m_i} \sum_j k_{ij} q_j = 0$$

Assume that $q_i = q_{i0} \sin(At + B)$, implying coherent move;

$$-A^2 q_{i0} \sin(At + B) + \frac{1}{m_i} \sum_j k_{ij} q_{j0} \sin(At + B) = 0$$

$$-\frac{A^2}{\lambda} \frac{q_{i0} \sqrt{m_i}}{\eta_i} + \sum_j \frac{k_{ij}}{\underbrace{\sqrt{m_i m_j}}_{b_{ij}}} \frac{q_{j0} \sqrt{m_j}}{\eta_j} = 0$$

Diagonalization

$$\sum_j b_{ij} \eta_j = \lambda \eta_i$$

In a matrix representation,

$$\mathbf{B} \vec{\eta} = \lambda \vec{\eta}$$

where $\mathbf{B} = [b_{ij}]$ and $\vec{\eta} = (\eta_1, \eta_2, \dots, \eta_N)^T$.

After diagonalization ($\det|\mathbf{B} - \lambda \mathbf{I}| = 0$),

- eigenvalues: normal-mode frequencies
- eigenvectors: normal-mode coordinates

